Yuan Shan

Personal website: **cedricshan.github.io**
M.S. in Statistical Science · Duke University

EDUCATION

Duke University Aug 2024 – Dec 2026 (expected)
M.S. in Statistical Science

Duke University Jul 2020 – Aug 2024
B.S. in Data Science

EXPERIENCE

Duke University — Graduate Research Assistant May 2026 – Present
Advised by Prof. Henry D. Pfister and Dr. Ziv Aharoni · [Showcase] [Interview] · First-author manuscript under review (2026)

- First-authored **StreetWise**, a two-tier vision-language system that gives blind and low-vision pedestrians proactive spoken hazard alerts from smart-glasses video; the 0.5B generator reaches 91% of GPT-5.6-terra's judged alert quality and surpasses every open-weight VLM evaluated (up to 38B) at **59 ms** time to first token.
- Designed the training pipeline: LoRA SFT of a **FastVLM-7B** teacher, sequence-level distillation into **FastVLM-0.5B**, then **Dr. GRPO** with a programmatic reward.
- Built **StreetWise-20K** (20k relabeled egocentric clips) and **StreetWise-Bench** (300 clips, VLM-judge rubric plus latency); reproduced the WAD GPT-Score and VIABench protocols.

Wenge Tech — LLM Algorithm Engineer Intern May 2025 – Oct 2025
[Paper] [Code & Benchmark] · Accepted at IEEE ICASSP 2026

- Developed **MEOW**, the first metadata-driven, end-to-end LLM framework for academic survey outline generation, combining supervised fine-tuning (SFT) and reinforcement learning (GRPO) within the RLHF paradigm.
- Designed and implemented a benchmark and multi-dimensional evaluation metrics (structure, content, pragmatics) for outline quality assessment; curated and distilled ∼20k+ survey datasets from arXiv, bioRxiv, and medRxiv.
- Constructed Chain-of-Thought (CoT) annotations to guide structured reasoning, materially improving logical taxonomy construction and overall coherence of generated survey outlines.
- Optimized **Qwen3-8B** model (full & LoRA) with SFT and reinforcement learning; designed reward functions (structural distance & format compliance) to align generation with human writing preferences.
- Leveraged **vLLM** to accelerate inference and RL training, achieving state-of-the-art performance, surpassing SurveyX and outperforming strong LLMs (e.g., GPT-5 Nano, DeepSeek-R1) on evaluation benchmarks.

Beijing Academy of Artificial Intelligence (BAAI) — Research Intern Jun 2023 – Oct 2023

- Contributed to the development of the 3D electron microscopy processing toolchain at BAAI, with a focus on advanced neuron segmentation methodologies such as **PyTorch Connectomics (PyTC)** and **Local Shape Descriptors (LSD)**.

Tencent — Machine Learning Engineering Intern Jun 2022 – Aug 2022

- Independently undertook a project to identify potential video categories on the Tencent Video platform, managing the entire process from data extraction, cleaning, and algorithm implementation through to the reporting of results.
- Developed machine learning models based on **XGBoost** to construct quantitative indicators for video potential; used SQL and Python for data retrieval and ML model implementation.
- Employed **Spark** for distributed computing to enhance pipeline efficiency with data volumes up to **7,500,000 records**.

SELECTED PROJECTS

Music Chat Recommender Apr 2026
[Live Demo] [GitHub]

- Built an interactive music-recommendation chatbot with **Gradio** as the front-end and **Groq Llama 3.3 70B** as the LLM backend, with multi-turn conversation memory so the assistant tracks user preferences across turns.
- Connected the chatbot to the **iTunes Search API** for track and artist metadata and the **YouTube Data API** for playable video links, returning recommended songs together with album art and direct YouTube embeds inside the chat.

Bradley–Terry Analysis of the English Premier League Mar 2026
[Slides] [GitHub]

- Fit a **Bradley–Terry** paired-comparison model — reformulated as logistic regression to enable standard MLE inference — to 21 seasons of English Premier League match data (2005/06–2025/26).
- Extended the base model with the **Davidson** formulation for draws plus season-specific and team-specific home-advantage parameters; validated each extension via Wald and likelihood-ratio tests against nested baselines.

HONORS

Dean's List with Distinction (DKU) · Finalist Prize, MCM/ICM · First Prize, National Olympiad in Informatics (NOI) in Provinces

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